

In [1]: `# related mostly to https://www.kaggle.com/phuongpm/mbti-prediction/data`

```
import pandas as pd # data processing, CSV file I/O (e.g. pd.read_csv)
import numpy as np # linear algebra
import sklearn

import re
import string

from sklearn.feature_extraction.text import CountVectorizer, TfidfTransformer, TfidfVectorizer
from sklearn.preprocessing import LabelEncoder
from sklearn.linear_model import LogisticRegression
from sklearn.model_selection import train_test_split, RandomizedSearchCV, GridSearchCV, StratifiedKFold
from xgboost import XGBClassifier, plot_importance
from sklearn.metrics import accuracy_score, roc_auc_score
from sklearn.feature_selection import SelectFromModel
from itertools import compress

from pylab import rcParams
import seaborn as sns
import matplotlib.pyplot as plt
from wordcloud import WordCloud, STOPWORDS

from nltk.corpus import stopwords
from nltk import word_tokenize
from nltk.stem import PorterStemmer, WordNetLemmatizer
```

In [3]: `data = pd.read_csv('input/kaggle_mbti.csv')
data.head()`

	type	posts
0	INFJ	'http://www.youtube.com/watch?v=qsXHcwe3krw ...
1	ENTP	'I'm finding the lack of me in these posts ver...
2	INTP	'Good one ____ https://www.youtube.com/wat...
3	INTJ	'Dear INTP, I enjoyed our conversation the o...
4	ENTJ	'You're fired. That's another silly misconce...

In [4]: `dist = data['type'].value_counts()
dist`

INFP	1832
INFJ	1470
INTP	1304
INTJ	1091
ENTP	685
ENFP	675
ISTP	337
ISFP	271
ENTJ	231
ISTJ	205
ENFJ	190
ISFJ	166
ESTP	89
ESFP	48
ESFJ	42
ESTJ	39
Name: type, dtype: int64	

In [5]: `dist.index`

Out[5]: `Index(['INFP', 'INFJ', 'INTP', 'INTJ', 'ENTP', 'ENFP', 'ISTP', 'ISFP', 'ENTJ', 'ISTJ', 'ENFJ', 'ISFJ', 'ESTP', 'ESFP', 'ESFJ', 'ESTJ'],
 dtype='object')`

In [6]: `data['seperated_post'] = data['posts'].apply(lambda x: x.strip().split("|||"))
data['num_post'] = data['seperated_post'].apply(lambda x: len(x))
data.head()`

	type	posts	seperated_post	num_post
0	INFJ	'http://www.youtube.com/watch?v=qsXHcwe3krw ...	['http://www.youtube.com/watch?v=qsXHcwe3krw, ...	50
1	ENTP	'I'm finding the lack of me in these posts ver...	['I'm finding the lack of me in these posts ve...	50
2	INTP	'Good one ____ https://www.youtube.com/wat...	['Good one ____ https://www.youtube.com/wa...	50
3	INTJ	'Dear INTP, I enjoyed our conversation the o...	['Dear INTP, I enjoyed our conversation the ...	50
4	ENTJ	'You're fired. That's another silly misconce...	['You're fired., That's another silly misconce...	50

In [7]: `# Before expanding the dataframe, give everyone an unique ID?
data['id'] = data.index
data.head()`

	type	posts	seperated_post	num_post	id
0	INFJ	'http://www.youtube.com/watch?v=qsXHcwe3krw ...	['http://www.youtube.com/watch?v=qsXHcwe3krw, ...	50	0
1	ENTP	'I'm finding the lack of me in these posts ver...	['I'm finding the lack of me in these posts ve...	50	1
2	INTP	'Good one ____ https://www.youtube.com/wat...	['Good one ____ https://www.youtube.com/wa...	50	2
3	INTJ	'Dear INTP, I enjoyed our conversation the o...	['Dear INTP, I enjoyed our conversation the ...	50	3
4	ENTJ	'You're fired. That's another silly misconce...	['You're fired., That's another silly misconce...	50	4

In [8]: `expanded_df = pd.DataFrame(data['seperated_post'].tolist(), index=data['id']).stack().reset_index(level=1, drop=True).reset_index(name='idposts')
expanded_df.head()`

	id	idposts
0	0	'http://www.youtube.com/watch?v=qsXHcwe3krw
1	0	http://41.media.tumblr.com/tumblr_ifouy03PMA1q...
2	0	enfp and intj moments https://www.youtube.com...
3	0	What has been the most life-changing experienc...
4	0	http://www.youtube.com/watch?v=vXZeYwwRDw8 h...

In [9]: `expanded_df=expanded_df.join(data.set_index('id'), on='id', how = 'left')
expanded_df=expanded_df.drop(columns=['posts','seperated_post','num_post'])
expanded_df.head()`

	id	idposts	type
0	0	'http://www.youtube.com/watch?v=qsXHcwe3krw	INFJ
1	0	http://41.media.tumblr.com/tumblr_ifouy03PMA1q...	INFJ
2	0	enfp and intj moments https://www.youtube.com...	INFJ
3	0	What has been the most life-changing experienc...	INFJ
4	0	http://www.youtube.com/watch?v=vXZeYwwRDw8 h...	INFJ

```
def clean_text(text):
    # Lemmatizer | Stemmatizer
    stemmer = PorterStemmer()

    # Cache the stop words for speed
    cachedStopWords = stopwords.words("english")

    text = re.sub(r'http[^s]*', '', text)
    text = re.sub('[0-9]+', '', text).lower()
    text = re.sub('[a-z0-9]+', 'user', text)
    text = re.sub("[a-zA-Z]", " ", text) #keep only words

    # some additional info about this that https://towardsdatascience.com/fuzzy-string-match-with-python-on-large-dataset-and-why-you-should-not-use-fuzzywuzzy-4ec9f0defcd
    stemmer.stem(" ".join([w for w in text.split(' ') if w not in cachedStopWords]))

    text = re.sub(' +', ' ', text).lower() # Remove spaces > 1
    text = re.sub('[%s]*' % string.punctuation, '', text)

    return text
```

In [11]: `final_df = expanded_df.copy()`

In [12]: `final_df['idposts'] = final_df['idposts'].apply(clean_text)`

In [13]: `final_df.head()`

	id	idposts	type
0	0		INFJ
1	0		INFJ
2	0	enfp and intj moments sportscenter not top ten...	INFJ
3	0	what has been the most life changing experienc...	INFJ
4	0	on repeat for most of today	INFJ

In [14]: `cleaned_df = final_df.groupby('id')['idposts'].apply(list).reset_index()
cleaned_df.head()`

	id	idposts
0	0	[, , enfp and intj moments sportscenter not t...
1	1	[i m finding the lack of me in these posts ve...
2	2	[good one , of course to which i say i know t...
3	3	[dear intp i enjoyed our conversation the oth...
4	4	[you re fired , that s another silly misconce...

In [15]: `data['clean_posts'] = cleaned_df['idposts'].apply(lambda x: ' '.join(x))
data.head()`

	type	posts	seperated_post	num_post	id	clean_posts
0	INFJ	'http://www.youtube.com/watch?v=qsXHcwe3krw ...	['http://www.youtube.com/watch?v=qsXHcwe3krw, ...	50	0	enfp and intj moments sportscenter not top ...
1	ENTP	'I'm finding the lack of me in these posts ver...	['I'm finding the lack of me in these posts ver...	50	1	i m finding the lack of me in these posts ver...
2	INTP	'Good one ____ https://www.youtube.com/wat...	['Good one ____ https://www.youtube.com/wa...	50	2	good one of course to which i say i know tha...
3	INTJ	'Dear INTP, I enjoyed our conversation the o...	['Dear INTP, I enjoyed our conversation the ...	50	3	dear intp i enjoyed our conversation the othe...
4	ENTJ	'You're fired. That's another silly misconce...	['You're fired., That's another silly misconce...	50	4	you re fired that s another silly misconcept...

```
def clean_text_extra_spaces(text):
    stemmer = PorterStemmer()

    text = re.sub(' +', ' ', text).lower() # Remove spaces > 1

    return re.sub('[%s]*' % string.punctuation, '', text)
```

In [17]: `data['clean_posts'] = data['clean_posts'].apply(clean_text_extra_spaces)
data.head()`

	type	posts	seperated_post	num_post	id	clean_posts
0	INFJ	'http://www.youtube.com/watch?v=qsXHcwe3krw ...	['http://www.youtube.com/watch?v=qsXHcwe3krw, ...	50	0	enfp and intj moments sportscenter not top te...
1	ENTP	'I'm finding the lack of me in these posts ver...	['I'm finding the lack of me in these posts ve...	50	1	i m finding the lack of me in these posts ver...
2	INTP	'Good one ____ https://www.youtube.com/wat...	['Good one ____ https://www.youtube.com/wa...	50	2	good one of course to which i say i know that...
3	INTJ	'Dear INTP, I enjoyed our conversation the o...	['Dear INTP, I enjoyed our conversation the ...	50	3	dear intp i enjoyed our conversation the othe...
4	ENTJ	'You're fired. That's another silly misconce...	['You're fired., That's another silly misconce...	50	4	you re fired that s another silly misconcepti...

In [18]: `data['E_I'] = data['type'].apply(lambda x: 1 if x[0] == 'E' else 0)
data['S_N'] = data['type'].apply(lambda x: 1 if x[1] == 'S' else 0)
data['T_F'] = data['type'].apply(lambda x: 1 if x[2] == 'T' else 0)
data['J_P'] = data['type'].apply(lambda x: 1 if x[3] == 'J' else 0)`

In [19]: `data.head()`

	type	posts	seperated_post	num_post	id	clean_posts	E_I	S_N	T_F	J_P
0	INFJ	'http://www.youtube.com/watch?v=qsXHcwe3krw ...	['http://www.youtube.com/watch?v=qsXHcwe3krw, ...	50	0	enfp and intj moments sportscenter not top te...	0	0	0	1
1	ENTP	'I'm finding the lack of me in these posts ver...	['I'm finding the lack of me in these posts ver...	50	1	i m finding the lack of me in these posts ver...	1	0	1	0
2	INTP	'Good one ____ https://www.youtube.com/wat...	['Good one ____ https://www.youtube.com/wa...	50	2	good one of course to which i say i know that...	0	0	1	0
3	INTJ	'Dear INTP, I enjoyed our conversation the o...	['Dear INTP, I enjoyed our conversation the ...	50	3	dear intp i enjoyed our conversation the othe...	0	0	1	1
4	ENTJ	'You're fired. That's another silly misconce...	['You're fired., That's another silly misconce...	50	4	you re fired that s another silly misconcepti...	1	0	1	1

In [20]: `data=data.drop(columns=['posts', 'seperated_post', 'num_post', 'id'])`

In [21]: `data.head()`

	type	clean_posts	E_I	S_N	T_F	J_P
0	INFJ	enfp and intj moments sportscenter not top te...	0	0	0	1
1	ENTP	i m finding the lack of me in these posts ver...	1	0	1	0
2	INTP	good one of course to which i say i know that...	0	0	1	0
3	INTJ	dear intp i enjoyed our conversation the othe...	0	0	1	1
4	ENTJ	you re fired that s another silly misconcepti...	1	0	1	1

In [22]: `data.to_csv('input/kaggle_mbti_data_clean.csv', index=False)`